

Trainings ▾

General Information

with Mechanically Actuated Injector

The design of the cylinder liner has changed since the introduction of the K38 and K50 engines.

The cylinder liner inside diameter is the same on all designs. The installed liner protrusion is the same for all designs. The liner protrusion for the K38 and K50 engines was changed. Refer to Procedure 001-064 in Section 1. (</qs3/pubsys2/xml/en/procedures/28/28-001-064-tr.html>) The depth of the counterbore on new blocks was changed to obtain the correct protrusion.

Revised brass seal rings with three locating tabs on the inside diameter were released in May, 1995. The tabs will hold the seal ring in the correct position on the cylinder liner during installation of the liner in the cylinder block, preventing incorrect assembly that can lead to coolant leaks at the liner counterbore.

The tabs are manufactured to have an interference fit to the lower press fit diameter of the cylinder liner, so the tabs will be positioned in the small radius immediately under the liner flange, holding the seal ring in position.

A revised rectangular sealing ring (crevice seal) was released for K38 and K50 engines in September, 1994, starting with Engine Serial Number (ESN) 33128708. The crevice seal is manufactured from nitrile material and provides improved durability with increased resistance to nibbling and erosion.

The revised and previous seals can be intermixed.

with Electronically Actuated Injector

Nitride hardened cylinder liners were introduced for engines with electronically actuated injectors, and these liners have the same dimensions as the liners used on engines with mechanically actuated injectors, but offer increased durability. They are gray in color as opposed to black.

Preparatory Steps



Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ WARNING ⚠

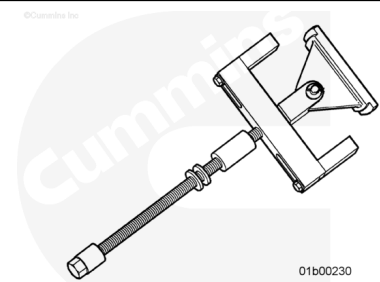
This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

- Disconnect the batteries. See equipment manufacturer service information.
- Remove the cylinder head. Refer to Procedure 002-004 in Section 2.
(/qs3/pubsys2/xml/en/procedures/28/28-002-004-tr.html)
- Remove the piston and connecting rod. Refer to Procedure 001-054 in Section 1.
(/qs3/pubsys2/xml/en/procedures/28/28-001-054-tr.html)

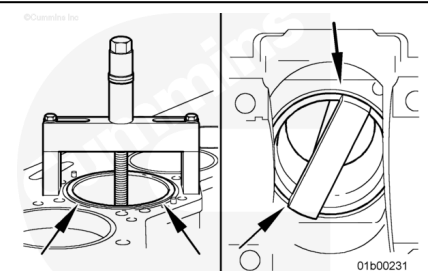
Remove

Use the cylinder liner puller, Part Number 3163745, and puller plate, Part Number 3162886, to remove the cylinder liner.

Wind down the threaded rod to extend the arm. Turn the arm to the side to enable passage through the liner.

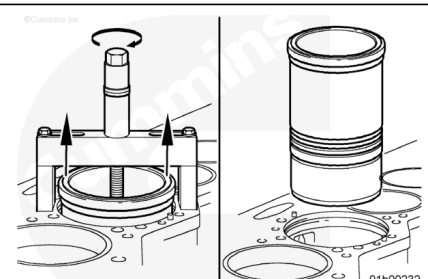


Install the cylinder liner puller, Part Number 3163745, in the cylinder liner. The puller feet **must not** touch the top of the liner. The puller arms **must** be positioned firmly on the bottom of the liner.



Turn the puller screw until the liner loosens in the block.

Remove the tool and the liner.

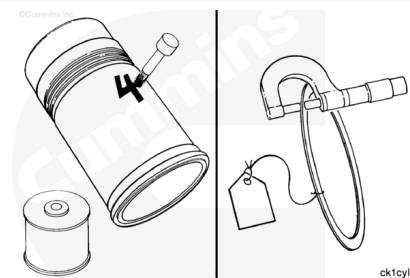


Use a liquid metal marker to mark the cylinder number on each liner.

If sealing rings were used:

- Use a tag to mark the cylinder number.
- Measure in several places and record the thickness of the sealing rings used in each cylinder.

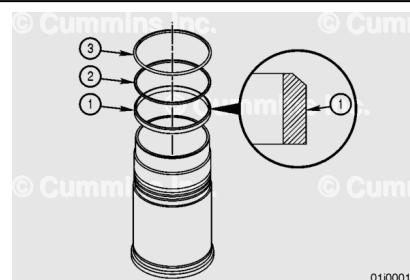
The thickness of the sealing ring is one factor in determining liner protrusion. This information **must** be known when the liners are installed in the engine.



Clean and Inspect for Reuse

Remove and discard the two d-rings (2 and 3).

Remove and discard the crevice seal (1).



⚠ WARNING ⚠

Wear appropriate eye and face protection when using a wire brush. Be sure the wire brush is rated for the correct rpm if the brush is motor driven. Flying debris and dirt can cause personal injury.



⚠ CAUTION ⚠

Do not use a hone, aluminum oxide cloth, or sandpaper to clean the cylinder liners. Abrasives can damage the finish and the pattern, and can contaminate the liner.

Use a high quality steel wire brush. Clean the liner flange seating area and the secondary press fit below the liner flange.

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

When using a steam cleaner, wear safety glasses or a face shield, as well as protective clothing. Hot steam can cause serious personal injury.

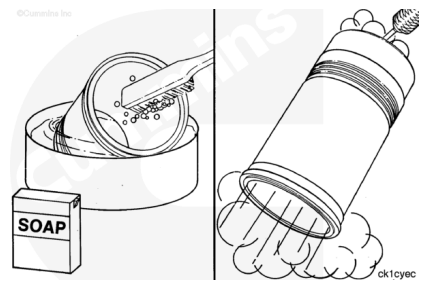
⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Use a brush that is **not** metallic, detergent soap, and warm water. Clean the inside diameter

Clean the liners with solvent or steam.

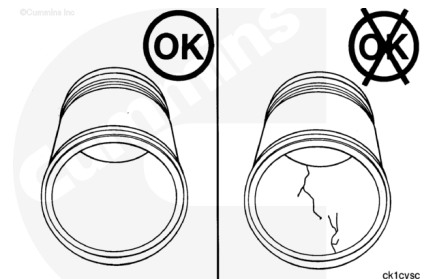
Dry with compressed air.



Note : See Parts Reuse Guidelines, Bulletin 3810303, for examples of liner condition.

Check the liners for cracks on the inside diameter and outside diameter.

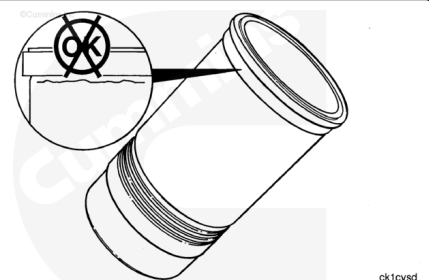
Any liner containing cracks **must** be replaced.



Check for cracks under the flange.

Note : Cracks can also be detected by either magnetic inspection or the dye method.

Any liner containing cracks **must** be replaced.

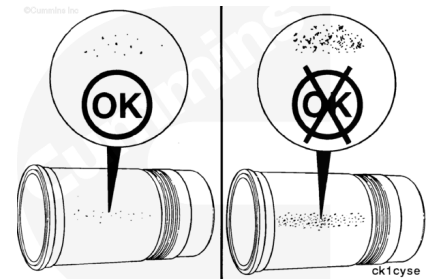


Check the outside diameter for excessive corrosion or pitting.

Pits **must not** exceed 1.6 mm [0.0625 in] in depth.

Replace the liner if the pits are too deep or if it is **not** possible to remove the corrosion with a fine emery cloth.

Note : If a liner that contains minimum pitting is installed again, it **must** be installed so that the pitting is in line with the centerline of the crankshaft.

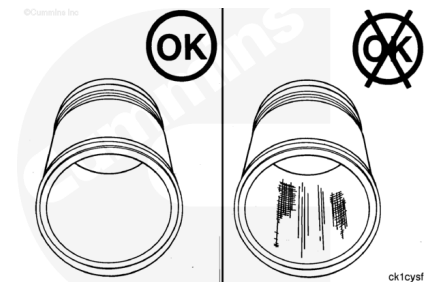


Check the inside diameter for vertical scratches deep enough to be felt with a fingernail.

Note : If the scratch is felt with a fingernail, the liner **must** be replaced.

Check the inside diameter for scuffing or scoring.

Any liner that contains scuffs or scores **must** be replaced.

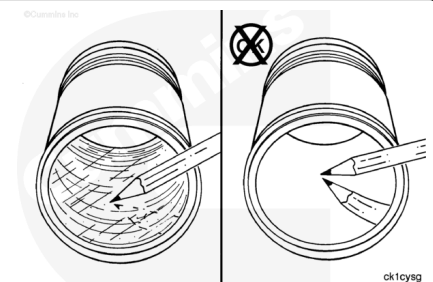


Check the inside diameter for liner bore polishing.

A light polish produces a bright mirror finish in the worn area with traces of the Lubrite coating remaining and the original hone marks remaining.

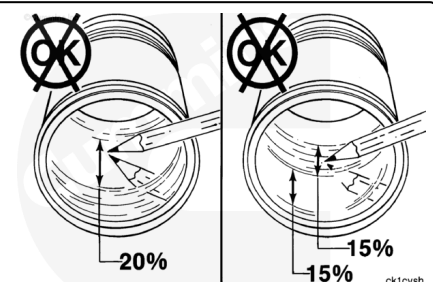
A moderate polish produces a bright mirror finish in the worn area with very light traces of the original hone marks or an indication of an etch pattern.

A heavy polish produces a bright mirror finish in the worn area with **no** traces of hone marks or an etch pattern.



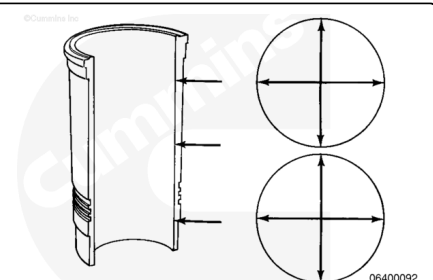
Replace the liner if:

- A heavy polish is present over 20 percent of the piston ring travel area.
- 30 percent of the piston ring travel area has both moderate and heavy polish, and half (15 percent) of the area has heavy polish.



Use a dial bore gauge. Measure the inside diameter of the liner at the:

- Top
- Bottom
- Middle of the piston ring travel area.



The middle of piston ring travel is approximately 100 mm [3.94 in] down the bore, as measured from the top of the cylinder liner. Record the cylinder liner inside diameter at this point, so that it can be used for piston ring gap measurement.

Take two measurements at each location. The measurements **must** be 90 degrees from the initial measurement.

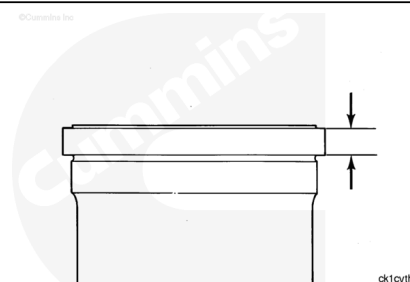
Cylinder Liner Inside Diameter		
mm		in
158.737	MIN	6.2495
158.877	MAX	6.2550

Note : The inside diameter of a new cylinder liner can be 0.015 mm [0.0006 in] smaller than specification because of the Lubrite coating.

Use a micrometer. Measure the thickness of the liner flange.

Note : To adjust the protrusion, use this measurement to predict liner protrusion. Refer to Procedure 001-058 in Section 1. (</qs3/pubsys2/xml/en/procedures/28/28-001-058-tr.html>)

Cylinder Liner Flange Thickness			
	mm		in
Standard	13.36	MIN	0.526
	13.39	MAX	0.527
0.25 mm [0.010 in]	13.61	MIN	0.536
	13.64	MAX	0.537
0.51 mm [0.020 in]	13.87	MIN	0.546
	13.89	MAX	0.547
0.76 mm [0.030 in]	14.12	MIN	0.556
	14.15	MAX	0.557
1.02 mm [0.040 in]	14.38	MIN	0.566
	14.40	MAX	0.567



Note : Counterbore rings that are oversize in thickness aid the adjustment of liner protrusion.

Counterbore Ring Thickness

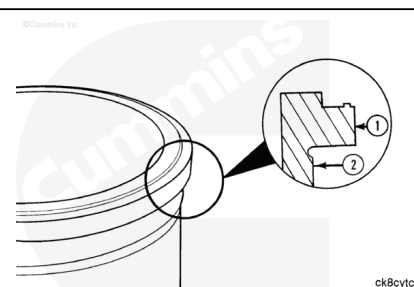
	mm		in
Standard	5.055	MIN	0.1990
	5.067	MAX	0.1995
0.05 mm [0.002 in]	5.105	MIN	0.2010
	5.118	MAX	0.2015
0.10 mm [0.004 in]	5.156	MIN	0.2030
	5.169	MAX	0.2035

Use a micrometer. Measure the outside diameter at the liner flange (1).

See the following table. Liner outside diameter specifications are the same for all K38 and K50 cylinder block designs.

Cylinder Liner outside diameter (1)

	mm		in
Standard	190.309	MIN	7.4925
	190.360	MAX	7.4945
0.25 mm [0.010 in] Oversize	190.563	MIN	7.5025
	190.614	MAX	7.5045
0.51 mm [0.020 in] Oversize	190.817	MIN	7.5125
	190.868	MAX	7.5145
0.76 mm [0.030 in] Oversize	191.071	MIN	7.5225
	191.122	MAX	7.5245
1.02 mm [0.040 in] Oversize	191.325	MIN	7.5325
	191.376	MAX	7.5345



Use a micrometer. Measure the outside diameter at the secondary press fit location (2).

See the following table. Secondary press fit outside diameter is the same for all K38 and K50 cylinder block designs.

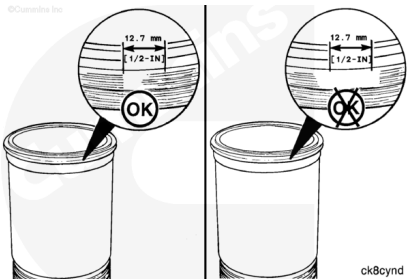
Cylinder Liner Outside Diameter (2)		
mm		in
181.81	MIN	7.158
181.86	MAX	7.160

Note : No oversize cylinder liners are oversize in the secondary press fit location.

Check the liner flange for wear. Wear on the outside diameter of the flange is indicated by small areas that are polished. The degree of wear is detected visually by the absence of machining marks.

If an area that does **not** have machining marks exceeds 13 mm [0.50 in] in length, the liner **must** be replaced.

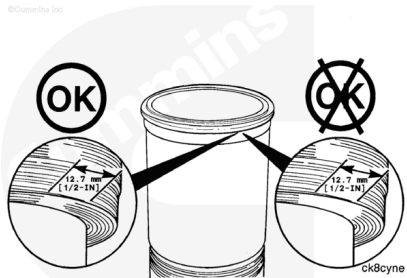
Note : If a liner that contains acceptable levels of small polished areas is installed again, it **must** be installed so that the polished area is in line with the centerline of the crankshaft.



Wear on the bottom of the flange is indicated by the absence of machining marks in the polished area.

If the area that does **not** have machining marks exceeds 13 mm [0.50 in] in length, the liner **must** be replaced.

Note : If a liner that contains acceptable levels of small polished areas on the bottom of the liner's flange is installed again, it **must** be installed so that the polished area is 90 degrees to the centerline of the crankshaft. That is, the **not** polished area **must** be in line with the centerline of the crankshaft.

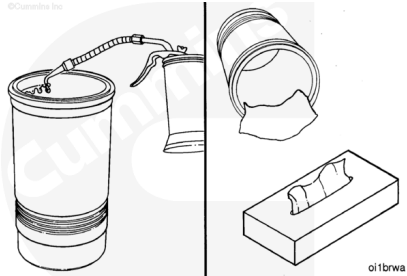


⚠ CAUTION ⚠

Do not set the liners in an area where dirty air flow can contaminate the liners.

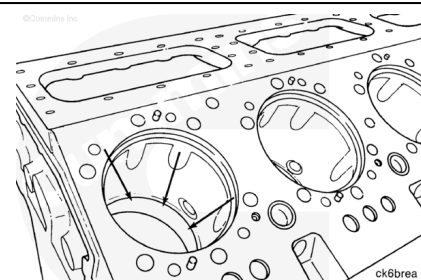
Apply a thick film of clean 15W-40 oil to the bores of the liners for final cleaning. Let the oil remain for 5 to 10 minutes.

Use a clean, lint-free paper towel, **not** cloth, to wipe the oil from the bores until the black and gray deposits are removed.



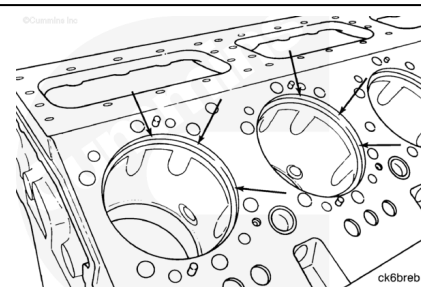
Install

Lubricate the packing ring bores of the cylinder block with vegetable oil.



⚠ WARNING ⚠

Do not use starting fluid. Naptha and methyl ketone are highly flammable and explosive. Follow the instructions of the manufacturer.

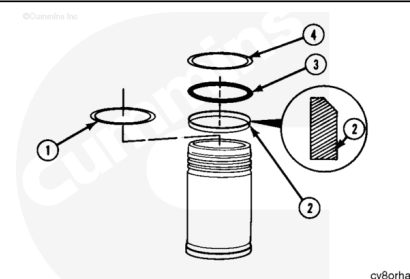


Use a hydrocarbon solvent, such as naptha, methyl ethyl ketone, trichlorethane 1,1,1 (methyl chloroform), or equivalent to clean the counterbore.

Note : The counterbore ledge must be dry. The sealant applied in assembly will not dry if there is oil on the ledge.

Note : The counterbore ledge must be dry. The sealant applied in assembly will not dry if there is oil on the ledge.

Install the shims (1) if required. The following table lists the part numbers and thicknesses.



Shims with Tabs

Part Number	mm	in	Number of Notches
3088385	0.46	0.018	two
3088386	0.52	0.0205	two
3088387	0.56	0.022	two
3088388	0.79	0.031	two

Due to the interference fit of the shims with tabs, it is necessary to keep the shim straight on the liner and push with your fingers near the tabs to work down over the lower press

fit diameter. Do **not** push on the shim too hard or the tabs can be deformed to the point that the seal will **not** fit squarely on the bottom of the liner flange. The thickest seal rings can deform to a very slight angle during installation on the liner. Some deformation is acceptable, as long as the tabs will still engage the radius under the liner flange.

Note : Do not lubricate the crevice seal and the d-ring seals or shims. This prevents oil transfer to the cylinder counterbore ledge during assembly.

The packing ring bore **must** be lubricated thoroughly.

Note : A new rectangular crevice seal was introduced in September 1994, ESN first 33128708), manufactured from nitrile material and provides improved durability and increased resistance to nibbling and erosion.

The beveled edge of the crevice seal (2) **must** be positioned as illustrated. Install the crevice seal.

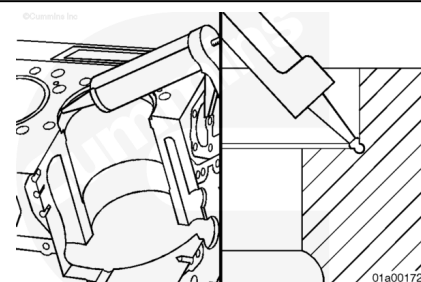
Note : The ring seals have a d-shaped cross section. This type of ring seal **must** be installed with the flat side against the cylinder liner.

Install the D-ring seals as shown. Use the mold mark on the seals to check if the seal is twisted.

- Black D-ring (3)
- Red D-ring (4).

⚠ CAUTION ⚠

The parts must be installed within 5 minutes after applying the sealant. An excessive amount of sealant can cause engine damage.



Install a bead of gasket sealant, Part Number 3823494, or equivalent, 1.1 to 1.6 mm [0.0468 to 0.0625 in] wide onto the cylinder block counterbore surface.

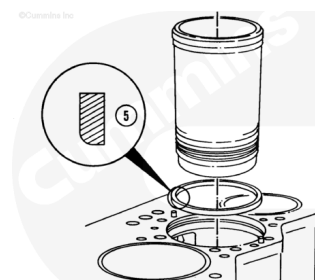
The sealant bead **must** be a continuous loop and **must not** contain any breaks or voids from start to finish.

Make sure that the end of the bead overlaps the beginning of the bead, approximately 6.0 mm [0.24 in].

Note : The sealant does not change the cylinder liner protrusion measurement. It prevents coolant leaks. The sealant fills the small spaces between the liner flange and the counterbore ring and block.

The counterbore ring **must** be positioned with the curved edge as illustrated. Install the counterbore ring (5).

Push the liner in the block by hand.



Use the liner installation tool, Part Number 3375422, or equivalent. Install the bridge assembly. Install two cylinder head capscrews.

Torque Value: 47 N•m [35 ft-lb]

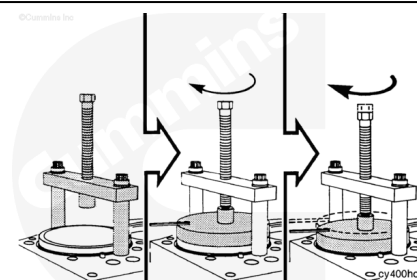
Install the push plate. Make sure the plate is aligned on the cylinder liner. Turn the push screw until it touches the push plate.

Use a wrench. Do **not** apply more than 68 N•m [50 ft-lb] of torque. Turn the push screw until the bottom of the liner flange touches the counterbore ring.

Remove the tool.

Check the liner protrusion. Use the following procedure for instructions on how to measure liner protrusion. Refer to Procedure 001-064 in Section 1.

(/qs3/pubsys2/xml/en/procedures/28/28-001-064-tr.html)



Finishing Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the piston and connecting rod. Refer to Procedure 001-054 in Section 1.
(/qs3/pubsys2/xml/en/procedures/28/28-001-054-tr.html)

- Install the cylinder head. Refer to Procedure 002-004 in Section 2.
(/qs3/pubsys2/xml/en/procedures/28/28-002-004-tr.html)
- Connect the batteries. See equipment manufacturer service information.
- Use the engine run-in procedure. Refer to Procedure 014-004 in Section 14.
(/qs3/pubsys2/xml/en/procedures/28/28-014-004-tr.html)
- Operate the engine and check for proper operation.

Last Modified: 01-Jun-2017
